

Plutus:

4 Quant ques + 5 coding (1:30 hrs total)

1. Smallest N for 100 trailing 0's in N!
2. N mathematicians and k rooms, in how many ways they can arrange themselves ?
3. You and your friend are playing a game by tossing a fair coin. The game rules are as follows-

For each coin toss:

- If the result is **heads**, you gain 1 point and your friend loses 1 point.
- If the result is **tails**, your friend gains 1 point and you lose 1 point.

The game continues until one player accumulates **2 points**, at which point the game ends.

- The player who reaches 2 points wins the game, and the other player must pay the winner **\$10**.

Part 1:

What is the expected value of the game for both players when it starts?

Part 2:

At any point during the game, you have the option to increase the prize from **\$10** to **\$20**, but this change can only be made once.

What is the optimal strategy to maximize the expected value, and what is the expected value of the game when following this strategy?

4. There are numbers from 1 to n, you randomly choose a number and that is 50, what is the best guess for n?

Coding-

1. Given 2 arrays A and B, where A represents cost per unit fuel and B represents required fuel units at that index to go ahead. You can purchase and store the fuel for the future, what is the minimum cost to travel from index 0 to n-1.
2. There is a graph with n nodes in which k nodes are power nodes (when you visit these your power refills to P) given edges and power nodes return the minimum power such that someone can traverse the nodes from 1 to n.
3. Start from (1,1) and go till (n,m),... from (i,j) one can jump only to (i+x[i][j] , j) or (i,j+X[i][j]) return the minimum number of steps required to reach (n,m) if you can't reach then return -1.

4. Given a string return lexicographically smallest string that can be made by swapping adjacent chars (any number of times). You can only swap adj chars if their colors are not the same (color string was given).
5. Given P,Q return numbers in the range of P and Q whose factors are not repeating.

OLA

Section 1.

16 statistics ques + 14 Data Science ques (mean, distribution, coefficient, covariance, t-score, p-value, z-score, confusion matrix, regression models, decision tree are imp)

1 coding que (easy-medium level)